

TEOBPM Calibration Against SXS Ringdown Injections

bayRing automatic calibration workflow

May 17, 2026

1 Scope

This report records a TEOBPM calibration campaign for non-eccentric, non-precessing SXS simulations. The campaign family is **unspecified**; aligned-spin campaigns are constructed hierarchically from a nonspinning baseline.

2 Dataset

Training simulations	0
Validation simulations	0
Validation fraction	n/a
Maximum eccentricity	n/a

3 Local Fits

Local TEOBPM fits are generated under `local_fit_configs/` and are intended to estimate peak quantities, post-peak TEOBPM coefficients, and relative merger phases mode by mode.

Indexed jobs	n/a
Completed jobs	n/a
Failed jobs	n/a
Collected coefficient rows	n/a
Collection failures	n/a

4 Global Fits

The global-fit file is versioned with schema `bayRing.teobpm.global-fit.v1`. The selection rule is the lowest Bayesian information criterion among hierarchical candidate bases, with the aligned-spin basis restricted to spin corrections on top of the nonspinning layer.

Modes with fitted targets	0
Fitted mode-target pairs	0

5 Validation

Validated coefficients	0
Median absolute residual	n/a
Maximum absolute residual	n/a

No validation figures were available when this report was rendered.

6 Open Production Steps

The workflow is ready to run the expensive SXS downloads and local fits. A final science report should add the generated mismatch figures after the validation table has been produced from the completed local-fit campaign.